

MT 304: Mechanical Behaviour of Materials

Course Code: MT 304

Course Name: Mechanical Behaviour of Materials

L-T-P-C: 3-0-2-4

Prerequisites: None

Intended for: B.Tech.

Mutual Exclusion: EN 611 - Durability Behavior of Energy Materials

Approval: 57th BoA; 64th BoA (EN 611 Durability Behaviour of Energy Materials discontinued and replaced by MT 304)

Preamble

The course examines the durability behavior of materials employed in various engineering applications. The challenge to durability comes from the deformation at room and elevated temperature under static and dynamic loading prevailing under a given application circumstance. In addition, corrosion, wear, and erosion limit the life of components. The course offers a holistic view of durability, indicating the directions of materials development for structural, thermal, and energy applications.

Course Outline

The course will start with a test of deformation behavior to identify the material parameters governing durability under static load and then proceed to understand the instantaneous deformation behavior under static and dynamic loading and after that, focuses on high-temperature time-dependent deformation behavior prevalent in high-temperature to follow the development of alloys for ultra-supercritical boilers and turbines. At the end the course focuses on durability threats to components employed in energy generation from corrosion, hot corrosion, erosion, and cavitation, as well as the strategies to counter them. The course will be delivered in the following modules.

Course Modules

- **Module 1:** Static loading and Deformation behavior Distinctive load – elongation behavior of different types of materials under uni-axial loading, definition of stress and strain, stress-strain diagram, elastic behavior, yielding and plastic behavior of ductile materials; necking instability, Important design criteria based on stress and strain at ambient temperature; strain rate and its influence of stress-strain diagram at elevated temperature. [4 Hours]
- **Module 2:** Durability under static loading Defects in materials; linear defects or dislocations – types, burgers vector, slip, slip planes and slip directions, cross slip and climb, movement of dislocations leading to plastic deformation, stacking fault and partial dislocations, strain hardening, grain boundaries, strengthening mechanisms; recovery, recrystallization and grain growth during heating of deformed materials. [4 Hours]
- **Module 3:** Durability under Creep Deformation, Deformation under static loading at elevated temperature - creep curve, mechanisms of creep, the temperature dependence of creep, deformation mechanism maps, cavitations, stress rupture versus creep, extrapolation schemes; Development of materials for ultra-supercritical boilers for application in furnace panels, super-heaters, thick section components, and steam lines. [4 Hours]

- **Module 4:** Durability under Cyclic Loading, Deformation under cyclic loading, high cycle fatigue – S-N curve, effect of mean stress, Miner rule, cyclic stress-strain curve, low cycle fatigue, strain life equation, effect of stress concentration and size of component, Design for fatigue, effect of temperature on fatigue; creep-fatigue interaction, development of turbine materials. [4 Hours]

Teaching Laboratory: Experiments

1. Tensile test on ferrous materials: Determine mechanical properties (yield stress, ultimate stress, breaking stress, percentage of reduction in area, percentage of elongation, and Young's modulus) for ferrous materials.
2. Tensile test on non-ferrous materials: Compare the stress-strain diagrams of non-ferrous materials with those of ferrous materials.
3. Compression test on brittle vs. ductile metallic materials: Analyze the mechanical properties of both brittle and ductile materials under compression.
4. Compression test on ceramic materials: Determine the mechanical properties of ceramic materials through compression testing.
5. Microstructure examination: Study and compare the grain structures in various metallic and non-metallic materials using optical or electron microscopy.
6. Creep test: Conduct a creep test to observe the time-dependent deformation behavior of a selected material under constant stress.
7. Stress corrosion cracking study: Investigate the effects of stress corrosion cracking in specific materials by applying tensile stress in a corrosive environment.
8. Hydrogen embrittlement study: Analyze the effects of hydrogen embrittlement on selected metallic specimens by subjecting them to hydrogen exposure.
9. Micro-hardness/nanoindentation test: Perform hardness tests on various materials using micro-hardness or nanoindentation techniques to assess material properties at small scales.
10. Fatigue test: Conduct fatigue tests on selected materials to evaluate their resistance to cyclic loading and determine fatigue life.
11. Impact resistance and hardness measurement: Measure the hardness and impact resistance of various samples to assess their toughness and durability.
12. Friction and wear study: Investigate the frictional behavior of different material combinations under varying loads, sliding speeds, temperatures, and lubrication conditions to predict the coefficient of friction and wear rates.

Text Books

1. George E. Dieter, *Mechanical Metallurgy*, McGraw Hill Book Company 1986
2. Meyers, M.A. and Chawla, K.K., *Mechanical Behavior of Materials*, 2nd edition, Prentice Hall, 2008.